

Input artefacts · Stages 1–2b

Per-function feature blocks

Assembly and output

Dalvik bytecode of v

invoke-virtual {v0}, L...
const-string v1, "..."
move-result-object v2
from classes*.dex · Stage 1

Function-call graph $G = (V, E)$



Suspicious-API tags and backward slices

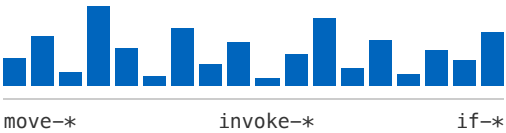
MobSF tagging · dangerous permissions —
Stage 2a
Sliced CFGs at each call site — Stage 2b

Method body of v

Syntactic paths through the decompiled
method, encoded by a pre-trained code2vec
model

3.1 Opcode histogram hop

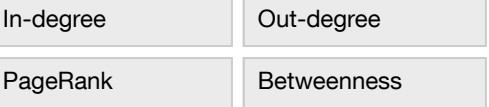
Frequency distribution of the Dalvik instructions in the body of v — raw one-hot counts, or TF-IDF weighted against the whole corpus to surface discriminative opcodes.



|0|

3.2 Structural metrics $Sstruct$

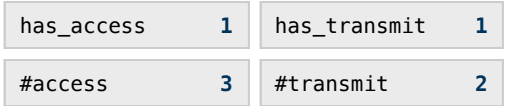
Position and influence of v inside the FCG. Standardised so that no metric dominates the learning process through its numerical range.



4

3.3 API flags and slice counts api

Direct, method-level indicators of the access-to-transmit exfiltration pattern: two binary flags plus the number of sensitive calls appearing in the slices associated with v .



4

3.4 Semantic embedding $c2v$

A dense 320-dimensional code2vec vector carrying the contextual meaning of the method — costlier than handcrafted features, but it recovers similarity across renamed, reused and repackaged code.



320

Concatenate

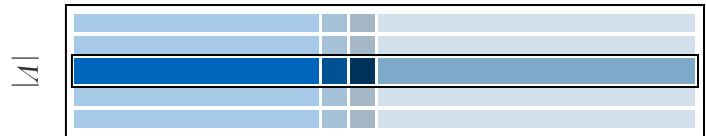


$$x_v = [hop \parallel Sstruct \parallel api \parallel c2v] \in \mathbb{R}^F$$

▼ z-score normalisation, per feature column

$$\hat{x} = (x - \mu) / \sigma \quad \text{comparable scales across heterogeneous blocks}$$

Node-feature matrix



$$X \in \mathbb{R}^{|V| \times F} \quad F = |0| + 4 + 4 + 320$$

Output

Serialised with the unified hyperedge set by the Hypergraph Constructor, Stage 6 — $\{H, W, X\}$.

The code2vec block is reused in **Stage 4B** for the semantic k -nearest-neighbour hyperedges ($\tau \approx 0.80$, $k = 5$).

Access and transmit evidence also weights the slice-aware hyperedges of **Stage 4C**.